IBNU BAGUS TRISNANDI

SENIOR SOFTWARE ENGINEER | SOLUTION ARCHITECT | TECHNICAL LEAD

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PROFESSIONAL SUMMARY

Senior Software Engineer and Solution Architect with 9+ years of experience delivering enterprise applications, AI platforms, IoT systems, analytics dashboards, and scalable backend services. Experienced in translating business requirements into secure technical architectures, leading multidisciplinary engineering teams, and managing end-to-end delivery from discovery and solution design through implementation and deployment. Strong expertise in full stack development, Retrieval-Augmented Generation (RAG), computer vision, event-driven integration, and on-premise AI infrastructure.

CORE COMPETENCIES

Architecture & Leadership: Solution Architecture, Technical Leadership, System Design, Requirements Analysis, SDLC, Agile Delivery, Team Mentoring
Software Engineering: JavaScript, TypeScript, Python, SQL, React, Vue, Node.js, Express.js, NestJS, REST APIs, Microservices
AI & Data: RAG, LLM Integration, Vector Databases, Computer Vision, Facial Recognition, Crowd Detection, Safety Hazard Detection, Agentic AI
IoT & Integration: MQTT, Kafka, Real-Time Data Pipelines, Sensor Integration, Monitoring Platforms, Event-Driven Architecture
Infrastructure & DevOps: Docker, CI/CD, Linux, AWS, GCP, On-Premise Servers, NVIDIA H200 GPU Infrastructure
Tools & Practices: Git, Figma, Code Review, Technical Documentation, Stakeholder Management

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Researcher, and Solution Architect | PT PLN Icon Plus, Jakarta **Sep 2019 - Present**

- Lead solution architecture and end-to-end engineering delivery for enterprise applications, AI initiatives, IoT platforms, and executive dashboards across the PLN Group ecosystem.
- Translate business objectives and operational challenges into scalable system designs, integration patterns, technical roadmaps, and implementation plans.
- Architect backend microservices and high-volume data pipelines using REST APIs, MQTT, Kafka, and containerized deployment patterns.
- Direct research and implementation of RAG, computer vision, chatbot, and on-premise AI solutions, including infrastructure planning for NVIDIA H200 GPU servers.
- Coordinate stakeholders, developers, infrastructure teams, and users throughout requirements discovery, development, testing, deployment, and continuous improvement.
- Mentor engineers, conduct code reviews, define technical standards, and improve software quality, security, scalability, and maintainability.

Software Developer | PT Solusi Teknologi Kreatif **Sep 2016 - Sep 2019**

- Developed custom web applications for commercial clients using React, Vue, Node.js, and modern API-based architectures.
- Managed the complete SDLC, including requirements gathering, solution design, development, testing, deployment, and maintenance.
- Built prototypes for IoT and emerging technology solutions to validate technical feasibility and business value.
- Collaborated with UI/UX designers and clients to deliver responsive, intuitive, and maintainable digital products.

Game Researcher, Virtual Reality | Telkom University **Jan 2016 - Apr 2016**

- Developed VR gameplay mechanics and 3D assets for academic research demonstrations and interactive prototypes.

SELECTED PROJECTS

Step Up Challenge PLN **2026**

Solution Architect and Developer

- Designed and developed a gamified employee engagement platform initiated by PLN General Affairs to encourage stair usage and reduce elevator load in the main office building.
- Created the event workflow, participation mechanism, progress tracking, leaderboard, and gamification concept to support measurable employee engagement.
- Integrated business rules and user experience requirements into a practical digital solution for campaign execution and monitoring.

Executive Dashboard, General Affairs PLN **2026**

Solution Architect and Developer

- Designed and developed an executive dashboard for centralized monitoring of five General Affairs functions and one consolidated executive view at PLN Head Office.
- Defined dashboard architecture, data visualization structure, cross-functional indicators, drill-down views, and monitoring workflows for management decision support.
- Worked with business stakeholders to convert operational requirements into clear metrics, data structures, and user-facing dashboards.

Smart and Green Building Platform PLN **2026**

Solution Architect and Developer

- Designed and developed a building monitoring dashboard integrating solar photovoltaic data and IoT-based building information.

- Enabled users to monitor energy performance, calculate efficiency indicators, and estimate avoided carbon emissions from renewable energy and efficiency initiatives.
- Defined the integration flow between field devices, data services, analytical calculations, and management dashboards.

AI CCTV Analytics for PLN **2026**

Solution Architect

- Designed the solution architecture for an AI-powered CCTV analytics platform supporting operational monitoring and workplace security across PLN environments.
- Defined computer vision use cases including facial recognition, crowd detection, restricted-area intrusion detection, and other configurable video analytics scenarios.
- Established high-level integration, processing, infrastructure, and dashboard requirements to support scalable deployment and centralized monitoring

AI Lab PLN Group **2025**

Solution Architect and Technical Lead

- Led the architecture and technical implementation of an AI experimentation lab for PLN Group employees to explore, test, and learn practical AI applications through guided workshops.
- Designed a modular environment supporting multiple use cases, including safety hazard detection, enterprise chatbot experimentation, and an interactive terminal playground.
- Defined the lab platform architecture, infrastructure requirements, demonstration workflows, technical standards, and workshop enablement approach.

AI PLN Mobile - RAG Customer Service Assistant **2025**

Solution Architect and Technical Lead

- Led development of a RAG-based customer service assistant integrated with the PLN Mobile application.
- Directed integration between the AI engine, retrieval layer, and mobile application backend to provide context-aware responses.

AI Legal Assistant - RAG Regulation Search **2024**

Solution Architect and Technical Lead

- Led research and development of an AI platform for semantic search across internal corporate regulations.
- Defined local LLM and NVIDIA H200 infrastructure requirements to support data privacy and secure enterprise deployment.

AI E-Arsip PLN - RAG Knowledge System **2024**

Solution Architect and Technical Lead

- Transformed a static document archive into an AI-powered knowledge platform using document chunking, embeddings, vector retrieval, and semantic search.
- Designed retrieval and knowledge-access patterns to improve the speed and usability of internal information discovery.

Battery Swapping Station PLN - End-to-End IoT Platform **2023**

Solution Architect and Researcher

- Built the end-to-end platform from hardware communication and firmware integration through backend services and operational dashboards.
- Enabled real-time station synchronization, battery health monitoring, and user transaction processing.

Power Monitoring System **2023**

Full Stack and IoT Engineer

- Developed a full-stack IoT dashboard using React and Express.js to visualize high-frequency electrical data and support proactive load monitoring.

HI+ Energy Dashboard **2021**

Full Stack Developer

- Built dashboards for energy monitoring, carbon footprint tracking, billing transparency, and green energy engagement.

FTTH Network Analysis Dashboard **2021**

Full Stack Developer

- Developed geospatial analytics to identify coverage gaps, demand areas, and network expansion opportunities.

ICONNet Website and Backoffice **2021**

Full Stack Developer

- Built customer-facing and internal administrative applications supporting customer registration and service workflows.

Additional Projects: Stroomnet Broadband Monitoring Dashboard (2020); AirSALE Regional Tax Dashboard (2020); AirSALE POS Web Application (2020); Employee Transfer System (2019); DoRun VR Game Prototype (2016); and other enterprise digitalization, AI, IoT, monitoring, and analytics initiatives across PLN Group.

EDUCATION

Master's Degree, Informatics Engineering, AI and Machine Learning | Universitas Gadjah Mada **2025 - Present**

Bachelor's Degree, Informatics Engineering | BINUS University **2021 - 2024**

Diploma, Informatics Engineering | Telkom University **2013 - 2016**